COHOMOLOGY WITH SUPPORT IN A CLOSED SUBSCHEME AGREES BETWEEN THE ZARISKI AND ETALE TOPOLOGIES FOR QUASI-COHERENT COEFFICIENTS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file etale-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the comparison of the long exact sequences for cohomology with supports, and the quasi-coherence argument for the higher direct images of the open immersion combined with the support triangle.

Editorial note. The label lemma-cohomology-with-support-quasi-coherent names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Editorial note. Pre-open gate: python operations/agent-authoring/tools/precard_gate.py –source etale-cohomology.tex –label lemma-cohomology-with-support-quasi-coherent –start 14155 –end 14187 –local-id 3567 returned RESULT=OK (L3567 free, occupancy FREE, public interval clear). Candidate taken from lane-s4/halfB-priority.json.

Editorial note. The statement and proof are copied from the pinned chapter file; the cited inputs are kept as external pointers with their canonical Stacks tags.

Allowed standard prerequisites (editorial).

- Agreement of Zariski, etale and fpqc cohomology for quasi-coherent modules. Same chapter, the theorem identifying the cohomology of the associated quasi-coherent sheaf on the small etale site with the usual cohomology of the module. Supplies the equalities H^q(X_etale, F^a) = H^q(X, F) and H^q(U_etale, F^a) = H^q(U, F) entering the long exact sequences for cohomology with supports.
- Quasi-coherence of the higher direct images along a quasi-compact open immersion. Cohomology of Schemes, the lemma that the higher direct images of a quasi-coherent module along a quasi-compact morphism are quasi-coherent. Supplies the quasi-coherence of R^q j_* in the second part of the lemma.
- Higher direct images commute with the associated quasi-coherent sheaf. Descent, the lemma comparing the higher direct images of the associated quasi-coherent sheaf on the small etale site with the associated sheaf of the higher direct images. Supplies the identification of i_*H^q_Z(F^a) with the associated sheaf of i_*H^q_Z(F) in the retrocompact case.
- The triangle relating cohomology with support to the cohomology of the open complement. Same chapter, the lemma producing the distinguished

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triangle computing cohomology with support in a closed subscheme. Supplies the final step translating the two identifications into the statement about the sheaves with support.

Source reference. Original lemma 01XJ: sources/stacks-chow/tags.txt, source lines 2520–2520. Prerequisite pointer invoked at proof line(s) 14181; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 071N: sources/stacks-chow/tags.txt, source lines 8997–8997. Prerequisite pointer invoked at proof line(s) 14184; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0A45: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 14072–14072. Prerequisite pointer invoked at proof line(s) 14186; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 03OG: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 1332–1332. Prerequisite pointer invoked at proof line(s) 14161; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 03P2: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 2280–2280. Prerequisite pointer invoked at proof line(s) 14176; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let X be a scheme. Let $Z \subset X$ be a closed subscheme. Let $\mathcal{F}$ be a quasi-coherent $\mathcal{O}_X$-module and denote $\mathcal{F}^a$ the associated quasi-coherent sheaf on the small étale site of X (Proposition 03OG). Then*

1. *$H^q_Z(X, \mathcal{F})$ agrees with $H^q_Z(X_{étale}, \mathcal{F}^a)$,*
2. *if the complement of Z is retrocompact in X, then $i_*\mathcal{H}^q_Z(\mathcal{F}^a)$ is a quasi-coherent sheaf of $\mathcal{O}_X$-modules equal to $(i_*\mathcal{H}^q_Z(\mathcal{F}))^a$.*

Proof. Let $j : U \to X$ be the inclusion of the complement of Z. The statement (1) on cohomology groups follows from the long exact sequences for cohomology with supports and the agreements $H^q(X_{étale}, \mathcal{F}^a) = H^q(X, \mathcal{F})$ and $H^q(U_{étale}, \mathcal{F}^a) = H^q(U, \mathcal{F})$, see Theorem 03P2. If $j : U \to X$ is a quasi-compact morphism, i.e., if $U \subset X$ is retrocompact, then R^qj_* transforms quasi-coherent sheaves into quasi-coherent sheaves (Cohomology of Schemes, Lemma 01XJ) and commutes with taking associated sheaf on étale sites (Descent, Lemma 071N). We conclude by applying Lemma 0A45. □